Robotics Ops Eng. - Home assignment

1 Introduction

This assignment is a compact, end-to-end slice of the Robotics Operations Engineer role: bring a humanoid up in simulation, drop it into a task it has never seen, and show - with data - that it can do the job reliably and reproducibly.

Concretely, you'll bring up a Unitree G1 in MuJoCo, procedurally generate a maze from a random seed, navigate the robot from start to goal on its own sensing, record a multi-sensor dataset of the run, and analyse that data against the operational KPIs in §3.

> Yes, use Claude Code - and any AI tool you like. We assume you will. That is exactly why the bar is judgment, not typing speed: held-out seeds you can't memorise, data integrity you have to verify, and a live demo.

2 Tasks

1 Reproducible setup

One command to stand up the whole environment - MuJoCo, the G1 model, deps - pinned and documented. A script or container is ideal. The goal: we run it once and it just works, on a clean machine, deterministically.

2 Seeded maze generator

A procedural maze built as MuJoCo geometry, parameterised by an integer `--seed` and a difficulty knob (grid size / corridor width). The same seed must reproduce the exact same maze; a new seed gives a new maze every run. Corridors must physically fit the G1's footprint. Emit the ground-truth solution path so path-efficiency can be scored.

3 Navigation

Drive the G1 from start to goal: perceive walls, plan, and move. Closed-loop on the robot's *own* sensing - see the rule below. You may command velocities to a pretrained walking policy (e.g. `unitree_rl_gym` sim2sim).

4 Data collection script

Record the run: head RGB-D, IMU, base/joint state, planned path and map, plus ground-truth pose *for scoring only*. Multi-rate streams, per-sample timestamps, a documented schema, and crash-safety - a `kill -9` mid-run must not corrupt the dataset. Hit and report a sustained capture-rate target.

5 Analysis & KPI report

Run a batch of held-out seeds, compute the KPIs in §3, and produce a report (notebook or HTML) with plots and a one-to-two-page writeup: design, tradeoffs, dominant failure mode, and the verdict - *"would you trust this robot to run unsupervised?"*

> Hard rule - no peeking. The navigation stack may <u>not</u> consume MuJoCo ground-truth pose. GT is reserved for *scoring* localization and trajectory error only. Localize from the robot's own sensors. We will check, and we'll ask about it in the live demo.

3 KPIs & metrics

Report a sensible subset of the metrics below. Present each as a distribution over seeds rather than a single point estimate, and report tuned and held-out seeds separately.

SUCCESS

Held-out solve rate

Reached goal within time limit	% of unseen seeds
Reported as a rate over N≥20 seeds	with 95% CI
Seen-vs-held-out gap	overfit check

MISSION

Efficiency & speed

Path efficiency	traveled ÷ optimal
Time-to-goal	median sim-s
Final goal error	m from center

SAFETY & MOTION

Did it move cleanly?

Wall collisions	contacts / run
Min wall clearance	m, distribution
Stuck events + recoveries	count, recover-time
Smoothness	heading-rate / jerk

LOCALIZATION

Did it know where it was?

ATE vs ground-truth	m, RMSE
Drift	% of distance
Map coverage	explored ÷ true free

DATA QUALITY · OPS CORE

Is the dataset trustworthy?

Achieved fps vs target	per stream
Frame-drop rate	% per stream
Inter-sensor sync error	ms, max skew
Schema completeness	% valid, 0 NaN/inf
Determinism	seed → identical run?
Crash-safety	valid after SIGKILL?

RELIABILITY & GENERALIZATION

Would it run unsupervised?

MTBF	meters / failure
Recovery success rate	% of stuck events
Solve-rate vs complexity	curve
Failure taxonomy	stuck / hit / lost / timeout
Real-time factor	sim-speed, CPU/RAM

4 The live demo

During the interview, we will provide a random seed you have not run before. You will generate the corresponding maze live and have the robot solve it as we observe, ideally with your telemetry and KPI view updating in real time.

ON SCREEN, LIVE

Solve	reached goal? y/n
Time-to-goal	live clock
Collisions / stuck	running count
Capture health	fps · drops · sync

```
# the live demo should be one command
make demo SEED=$RANDOM     # generate fresh maze → run → live KPI view
```

5 Deliverables

- Git repo with one-command setup and a README a teammate can follow.
- Runnable entry points - generate a maze by seed, run a navigation episode, collect data, reproduce the KPI report.
- KPI report - notebook or HTML with plots across held-out seeds.
- Live-demo command - `make demo SEED=<n>` or equivalent.

6 Stretch goals

Optional. These are how strong candidates separate themselves - pick one you'd enjoy defending.

- Online SLAM with loop closure instead of dead-reckoning.
- Live monitoring dashboard streaming KPIs during the run.
- Dynamic obstacles - a moving wall or a second agent.
- Multi-robot - two G1s in one maze, no collisions.